

Vanilla Just Isn’t Exotic Enough: Vol-Arbitrage and Exotic Pricing on AETHER_CRYSTAL

IMC Prosperity Round 4 — Manual Trading Analysis (revised)

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Abstract

The simulator generates Aether Crystal under geometric Brownian motion with zero drift and annualised volatility $\sigma = 251\%$, yet every quoted option implies $\sigma_{IV} \approx 212\%$. The 39 vol-point gap is the source of all edge. We price the eleven instruments with closed-form Black–Scholes for vanillas and Monte Carlo for the three exotics (chooser, binary put, knock-out put), verify pricing identities (Rubinstein chooser decomposition; reflection-principle barrier formula), and solve a linear program for the maximum-EV portfolio under volume constraints. The recommended position — buy 50 contracts of every vanilla, the chooser, and the binary put — has expected PnL \$2.87M with standard deviation across the 100-simulation scoring of \$1.65M, giving a 97% probability of a positive score. Crucially, the seemingly-attractive knock-out put short (\$28,500 expected) is **rejected** after a left-tail correlation analysis: the KO position is positively correlated with the rest of the portfolio (Corr = +0.107), so its tail blow-ups occur on the same paths where the long-vol book is already underperforming. Including it improves the mean by \$27k but worsens the q_{01} score by \$48k and the absolute worst case by \$465k — a poor reward-to-risk trade-off.

1 Market data and parameters

The underlying obeys

$$dS_t = \sigma S_t dW_t, \quad \sigma = 2.51, \quad S_0 = 50, \quad r = 0,$$

discretised at 4 steps per trading day with 252 trading days per year. We define $T_2 = 14/252 \approx 0.0556$ and $T_3 = 21/252 \approx 0.0833$. Note $\sigma\sqrt{T_3} = 0.7246$ — a regime where $N(d_1) \neq N(d_2)$ even at the money, so the standard “ATM call delta ≈ 0.5 ” intuition breaks down.

contract	K	type	T	bid	ask	BS theo	implied σ_{IV}
AC_50_P	50	put	T_3	12.00	12.05	14.143	2.121
AC_50_C	50	call	T_3	12.00	12.05	14.143	2.121
AC_35_P	35	put	T_3	4.33	4.35	5.870	2.122
AC_40_P	40	put	T_3	6.50	6.55	8.299	2.125
AC_45_P	45	put	T_3	9.05	9.10	11.072	2.119
AC_60_C	60	call	T_3	8.80	8.85	11.025	2.127
AC_50_P_2	50	put	T_2	9.70	9.75	11.631	2.089
AC_50_C_2	50	call	T_2	9.70	9.75	11.631	2.089

Table 1: Vanilla quotes, theoretical Black–Scholes values at $\sigma = 2.51$, and implied volatility from mid quotes.

Implied volatility is uniformly $\sigma_{IV} \approx 2.12$, against the true $\sigma = 2.51$. The gap is consistent across strikes and expiries, with the slightly lower IV at T_2 reflecting tiny variance loadings rather than true skew. *This is real edge, not a modelling error.*

2 Vanilla pricing

Standard Black–Scholes with $r = 0$:

$$C(S, K, T, \sigma) = S N(d_1) - K N(d_2), \quad P(S, K, T, \sigma) = K N(-d_2) - S N(-d_1),$$

where $d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$. Per-contract expected PnL on the buy side is $\mathbb{E}[\pi_{\text{buy}}] = 3000 \cdot (\mathbb{E}[\text{payoff}] - \text{ask})$, ranging from \$4,539 (AC_35_P) to \$6,533 (AC_60_C). All eight vanillas are buy-side +EV.

3 Exotic pricing

3.1 Chooser option AC_50_CO

The buyer chooses at T_2 whichever of call or put is in-the-money, then settles at T_3 :

$$\Pi_{\text{CO}} = \mathbf{1}_{S_{T_2} > K} \cdot \max(S_{T_3} - K, 0) + \mathbf{1}_{S_{T_2} \leq K} \cdot \max(K - S_{T_3}, 0).$$

By Rubinstein’s chooser identity (with $r = 0$):

$$V_{\text{CO}} = C(S_0, K, T_3) + P(S_0, K, T_2)$$

Substituting: $V_{\text{CO}} = 14.143 + 11.631 = 25.774$. Monte Carlo agrees to within 0.05. The bid is 22.20, ask is 22.30, giving edge $25.77 - 22.30 = \$3.47$ per unit, i.e. \$10,356 per contract — the largest single-trade edge in the round.

3.2 Binary put AC_40_BP

Pays \$10 if $S_{T_3} < 40$:

$$V_{\text{BP}} = 10 \cdot \Pr(S_{T_3} < 40) = 10 \cdot N(-d_2), \quad d_2 = \frac{\ln(50/40) - \frac{1}{2}\sigma^2 T_3}{\sigma\sqrt{T_3}} = -0.0545,$$

giving $V_{\text{BP}} = 10 \cdot N(0.0545) = 5.217$. Edge over ask \$5.10 is \$0.117 per unit (\$351 per contract). The position is *bounded* — max loss per contract is $5.10 \cdot 3000 = \$15,300$. Worth including.

3.3 Knock-out put AC_45_KO — pricing

A down-and-out put with strike $K = 45$, barrier $B = 35$, expiry T_3 . Under GBM with $r = 0$ the closed-form (Reiner–Rubinstein) is

$$V_{\text{KO}} = P(S_0, K, T_3) - \left(\frac{B}{S_0}\right)^{2\lambda-2} P\left(\frac{B^2}{S_0}, K, T_3\right) - (K - B) \left[N(d_3) - \left(\frac{B}{S_0}\right)^{2\lambda-2} N(d_4) \right]$$

with $\lambda = 1 + (r - \frac{1}{2}\sigma^2)/\sigma^2 \approx 0.5$. Direct Monte Carlo gives $V_{\text{KO}} \approx \$0.128$ vs. ask \$0.175 (buy –EV) and bid \$0.150 (sell +EV by \$0.022 per unit, i.e. \$57 per contract, \$28,500 at full size 500). The barrier is breached in $\Pr(\min_t S_t < 35) = 69.2\%$ of paths, so most of the time the option is dead-on-arrival.

4 Portfolio optimisation

4.1 Linear-programming formulation

Variables $q_i \geq 0$ for each contract on its +EV side, with bounds $q_i \leq V_i$ (volume cap). Objective:

$$\max_{\mathbf{q}} \sum_i q_i \cdot 3000 \cdot \text{edge}_i.$$

With no risk constraint, the unconstrained LP returns $q_i = V_i$ for all +EV trades.

4.2 Why static delta-hedging is dominated

Adding a linear delta constraint $|\sum_i q_i \delta_i| \leq 0$ trims $q_{AC.50.C.2}$ from 50 to 21 (and uses 200 short underlying as hedge), yielding $\mathbb{E}[\text{PnL}] = \$2,696,278$ — a sacrifice of \$162k versus the unconstrained LP. But the gain in PnL standard deviation is only $\$16.7\text{M} \rightarrow \15.5M , a 7% reduction. This makes static delta-hedging strictly dominated: variance is overwhelmingly driven by gamma/convexity, not first-order delta. The intuition: at $\sigma\sqrt{T_3} = 0.7246$, the price distribution at expiry is so dispersed that linear delta is a near-irrelevant component of the option value’s path-dependence. **Recommendation: do not delta-hedge.**

4.3 The KO short trap: positive correlation kills the trade

The +\$28,500 EV from selling 500 KO puts is mathematically real, but its tail behaviour is correlated with the rest of the book in the *wrong* direction. We compute (over 500,000 simulated paths):

$$\text{Corr}(\Pi_{\text{KO short}}, \Pi_{\text{rest of book}}) = +0.107.$$

A negatively correlated KO position would offset losses elsewhere; a positive correlation *amplifies* them. Why does this happen?

Decomposition of the bad-KO paths. The KO short loses money when the path stays above \$35 throughout $[0, T_3]$ and ends below \$45 — i.e. when the underlying meanders without ever making a large excursion. These “low realised vol” paths are *also* the worst paths for the long-vol book, since every vanilla and the chooser is a long-vol position priced under $\sigma_{IV} \approx 2.12$ but valued at $\sigma = 2.51$. Without realised vol, the long-vol leg under-performs.

Conditional analysis. Conditioning on the 0.1% worst KO paths ($N = 500$):

quantity	conditional mean	unconditional mean
KO-short PnL	−\$14,188,483	+\$26,934
Rest-of-book PnL	−\$ 6,567,666	+\$2,873,387
Total portfolio PnL	−\$20,756,149	+\$2,900,321

The rest of the book underperforms its mean by \$9.4M precisely on the paths where the KO short is blowing up. This is the opposite of a hedge.

4.4 Tournament-score impact of including the KO short

The score is the average PnL over $M = 100$ underlying paths. A naive “100-path averaging smooths everything” argument would suggest the KO tail risk is harmless. The actual tournament-score statistics across 10,000 simulated tournaments tell a different story:

metric	with KO short	without KO short	Δ
mean score	\$2,900,528	\$2,873,774	+\$26,754
std of score	\$1,671,437	\$1,652,516	+\$18,921
q_{01}	−\$ 608,179	−\$ 560,188	−\$47,991
q_{05}	+\$ 301,697	+\$ 308,617	−\$ 6,919
worst case	−\$2,641,705	−\$2,176,915	−\$464,790
$\text{Pr}(\text{score} < 0)$	2.98%	2.90%	+0.08pp

Verdict: drop the KO short. The trade adds \$27k to the mean but degrades *every* downside metric: q_{01} worsens by \$48k, the absolute worst case worsens by \$465k, and $\text{Pr}(\text{score} < 0)$ ticks up. The reward-to-tail-risk ratio is unattractive, especially for a one-shot tournament where minimising the chance of catastrophic underperformance matters more than optimising the last basis point of expected value.

Why 100-path averaging fails to smooth. A single-path KO blow-up is up to \$14.8M. Across 100 paths, that single event contributes \$148k to the score average. The probability of at least one tail-KO event in 100 paths is approximately $1 - (1 - 0.001)^{100} \approx 9.5\%$. So nearly 1-in-10 tournaments will see at least one blow-up that drags the score down. Combined with the positive correlation in §4.3, this is enough to overwhelm the small per-contract edge.

5 Final recommendation

contract	action	qty	price	contract EV
AC_50_P	BUY	50	12.05 ask	\$6,233
AC_50_C	BUY	50	12.05 ask	\$6,288
AC_35_P	BUY	50	4.35 ask	\$4,539
AC_40_P	BUY	50	6.55 ask	\$5,215
AC_45_P	BUY	50	9.10 ask	\$5,874
AC_60_C	BUY	50	8.85 ask	\$6,533
AC_50_P_2	BUY	50	9.75 ask	\$5,662
AC_50_C_2	BUY	50	9.75 ask	\$5,571
AC_50_CO	BUY	50	22.30 ask	\$10,356
AC_40_BP	BUY	50	5.10 ask	\$333
AC_45_KO	NO TRADE	0	—	\$0
AC underlying	no trade	0	—	\$0
Total expected score				\$2,873,387
Score std (across 100-sim avg)				\$1,652,516
q_{05} score				\$308,617
Pr(score > 0)				97.1%

6 Lessons applied from the previous round

The Round 3 auction post-mortem identified four weaknesses: prior miscalibration, no fixed-point Nash check, no asymmetric-incentive analysis, and no minimax-regret evaluation. The analogous failure modes in this round are mitigated by:

1. **Pricing-model verification.** Black–Scholes vs. Monte Carlo agree to $< 0.1\%$ on all vanillas. The chooser identity $V_{CO} = C(K, T_3) + P(K, T_2)$ is verified analytically and by MC.
2. **Stress-testing the actual scoring rule.** Simulating 10,000 independent 100-path tournaments produces the true score distribution rather than relying on per-path PnL. Without this, the 49% per-path loss probability would look alarming; with it, the 97% positive-score probability emerges.
3. **Correlation-aware tail-risk gating.** Initial reasoning treated the KO short as “small EV, isolated tail risk.” The correlation analysis (§4.3) revealed $\text{Corr} = +0.107$ with the rest of the book. Tail risk is not isolated when the underlying-path conditioning is shared with the main position.
4. **Reward-vs-tail evaluation.** Explicitly compared score distributions with and without the KO short across 10,000 tournaments rather than relying on expected-value comparisons. The \$27k mean gain is overwhelmed by the \$48k–\$465k worsening of every downside percentile.
5. **Hedging-instrument cost-benefit analysis.** Evaluated static delta-hedging via underlying short and confirmed it sacrifices \$162k EV for only 7% standard deviation reduction — gamma is the dominant variance contributor at $\sigma\sqrt{T} \approx 0.7$.

Appendix A: Key code listings

Listing 1: Black-Scholes pricing and chooser identity verification.

```
1 import numpy as np
2 from scipy.stats import norm
3
4 S0, sigma = 50.0, 2.51
5 T2, T3 = 14/252, 21/252
6
7 def bs_call(S, K, T, s):
8     d1 = (np.log(S/K) + 0.5*s**2*T) / (s*np.sqrt(T))
9     return S*norm.cdf(d1) - K*norm.cdf(d1 - s*np.sqrt(T))
10
11 def bs_put(S, K, T, s):
12     d1 = (np.log(S/K) + 0.5*s**2*T) / (s*np.sqrt(T))
13     return K*norm.cdf(-(d1 - s*np.sqrt(T))) - S*norm.cdf(-d1)
14
15 # Rubinstein chooser identity (r=0):
16 # Chooser = Call(K, T_long) + Put(K, T_decision)
17 V_chooser = bs_call(S0, 50, T3, sigma) + bs_put(S0, 50, T2, sigma)
18 print(f"Chooser theo = {V_chooser:.4f}") # 25.7742
```

Listing 2: KO-short correlation analysis.

```
1 N = 500_000
2 rng = np.random.default_rng(20260427)
3 n_steps = 21*4; dt = 1/(252*4)
4 Z = rng.standard_normal((N, n_steps))
5 log_inc = -0.5*sigma**2*dt + sigma*np.sqrt(dt)*Z
6 S_path = S0 * np.exp(np.cumsum(log_inc, axis=1))
7 S_T2 = S_path[:, 14*4 - 1]
8 S_T3 = S_path[:, 21*4 - 1]
9 S_min = S_path.min(axis=1)
10
11 # KO-short and rest-of-book per-path PnL
12 ko_short = 500*3000*(0.150 - np.where(S_min<35, 0,
13                                     np.maximum(45 - S_T3, 0)))
14
15 rest = (
16     50*3000*(np.maximum(50-S_T3, 0) - 12.05) + # AC_50_P
17     50*3000*(np.maximum(S_T3-50, 0) - 12.05) + # AC_50_C
18     50*3000*(np.maximum(35-S_T3, 0) - 4.35) + # AC_35_P
19     # ... all other long-vol legs
20 )
21 print(f"Corr(KO_short, rest) = {np.corrcoef(ko_short, rest)[0,1]:+.4f}")
22 # Output: +0.1068 -- positive correlation kills the trade
```

Listing 3: Tournament-score simulation.

```
1 def tournament_score(positions, n_trials=10_000, n_paths=100):
2     scores = np.empty(n_trials)
3     for t in range(n_trials):
4         Z = rng.standard_normal((n_paths, n_steps))
5         p = S0 * np.exp(np.cumsum(-0.5*sigma**2*dt
6                                   + sigma*np.sqrt(dt)*Z, axis=1))
7         # ... evaluate each leg's per-path PnL ...
8         scores[t] = portfolio_pnl_per_path.mean()
9     return scores
10
11 # With vs without KO short
12 s_with = tournament_score(positions_with_ko)
13 s_no = tournament_score(positions_no_ko)
14 # Mean: +27k. q01: -48k. Worst case: -465k. --> drop the trade.
```